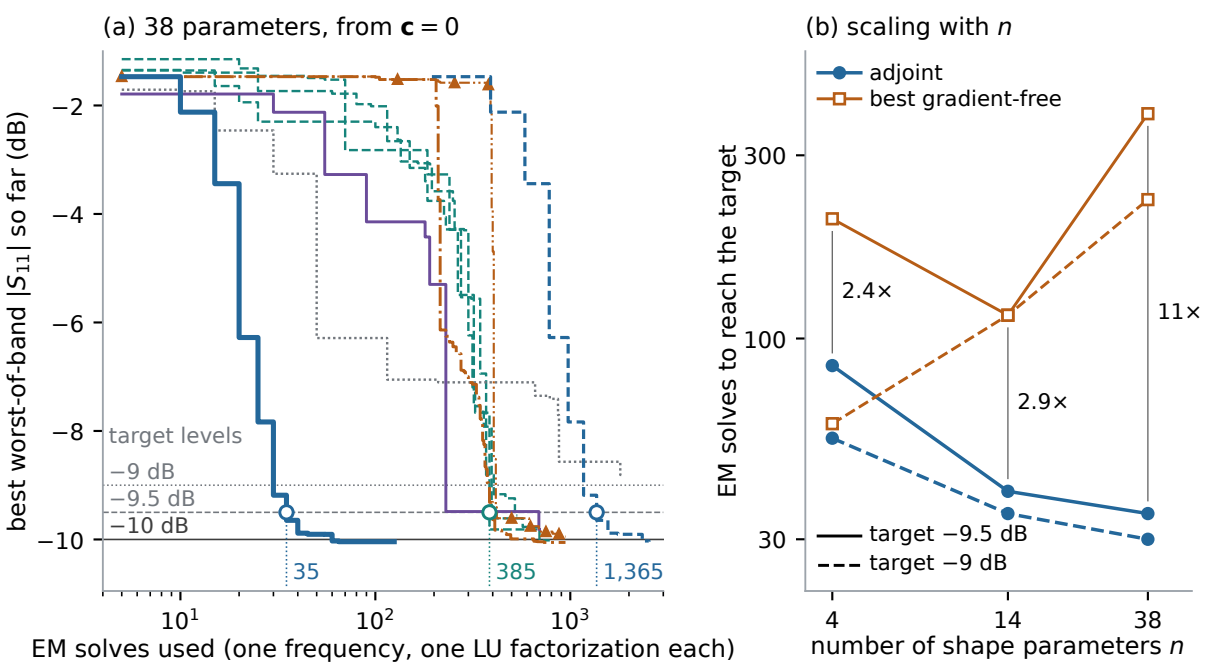
(a) 38 parameters, from c = 0
best worst-of-band |S11| so far (dB)
−2
−4
−6
−8
−10
target levels
−9 dB
−9.5 dB
−10 dB
35
385
1,365
10^1
10^2
10^3
EM solves used (one frequency, one LU factorization each)
(b) scaling with n
adjoint
best gradient-free
EM solves to reach the target
300
100
30
2.4×
2.9×
11×
target −9.5 dB
target −9 dB
4
14
38
number of shape parameters n
L-BFGS, adjoint gradient
L-BFGS, finite-difference gradient
sep-CMA-ES, σ0=0.5 (3 seeds)
CMA-ES (pycma), σ0=0.5
sep-CMA-ES, σ0=1.5 (1000 evals)
NEWUOA, npt 40
NEWUOA, npt 77 (Palace)